

Changes Evaluated in Soccer-Specific Power Endurance Either With or Without a 10-Week, In-Season, Intermittent, High-Intensity Training Protocol

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ABSTRACT

The purpose of this study was to evaluate changes in soccer-specific power endurance of 34 female high school soccer players throughout a season either with or without an intermittent, high-intensity exercise protocol. Thirty-four female high school soccer players were tested prior to the 2000 fall season and again 10 weeks later. The tests included an abridged 45-minute shuttle test (LIST), hydrostatic weighing, vertical jump, 20-m running-start sprint, and 30-second Wingate test. The experimental group (EG; $n = 17$, age 16.5 ± 0.9 years) completed a 10-week in-season plyometric, resistive training, and high-intensity anaerobic program. The control group ($n = 17$, age 16.3 ± 1.4 years) completed only traditional aerobic soccer conditioning. Statistical significance was set at $\alpha < 0.05$. The experimental group showed significant improvements in the LIST (EG = $\Delta 394$ seconds ± 124 seconds), 20-m sprint (EG = $\Delta -0.10$ seconds ± 0.10 seconds), increase in fat-free mass (EG = $\Delta 1.14$ kg ± 1.22 kg), and decreases in fat mass (EG = $\Delta -1.40$ kg ± 1.47 kg) comparing pre- to postseason. This study indicates that a strength and plyometric program improved power endurance and speed over aerobic training only. Soccer-specific power endurance training may improve match performance and decrease fatigue in young female soccer players.

Key Words: adolescents, fitness, tests, anaerobic, plyometric

Reference Data: Siegler, J., S. Gaskill, and B. Ruby. Changes evaluated in soccer-specific power endurance either with or without a 10-week, in-season, intermittent, high-intensity training protocol. *J. Strength Cond. Res.* 17(2):379–387. 2003.

Introduction

Soccer, when played at the elite level, requires tremendous endurance, speed, agility, and power. Research in this area has consistently focused on creating new techniques for enhancing the performance capa-

bilities of elite soccer players. These techniques have led investigators to create soccer-specific exercise testing protocols by extensively analyzing match play at the elite level (3, 22, 27). In doing so, investigators have defined soccer as an intermittent, high-intensity exercise. Fitness improvements specific to this activity pattern have further been defined as power endurance.

Studies have been conducted to monitor power endurance activity such as overall distance covered; average intensities; and percentage of time spent walking, jogging, running, and sprinting during match play (3, 21, 22, 27). These studies of movement patterns have provided researchers with the scientific basis for developing not only soccer-specific exercise testing procedures, but also applicable conditioning/training protocols (2, 16, 19, 20, 25). These conditioning protocols have been designed to focus on the high-intensity, intermittent aspects of fitness (11, 13, 26). Exercises include plyometrics, resistive training, and various forms of intermittent anaerobic training (12, 13, 26, 28).

The benefit of incorporating soccer-specific training into fitness protocols and assessment procedures has been well documented at the elite level (1, 4, 7, 9, 11–13, 20, 26, 28). However, a continual focus on elite players creates the problem of training protocol interpretation and testing procedures for the thousands of coaches and young athletes competing at the nonelite level. Consequently, training at the youth level has suffered. Exercise protocols at the youth level have traditionally focused on increasing aerobic endurance, but relatively little attention has been paid to the high-intensity aspect so crucial in soccer (7, 15). The purpose of this investigation was to study the soccer-specific changes in power endurance of female high school soccer players throughout the varsity season either with or without an intermittent, high-intensity training protocol.

Performance measures for youth soccer designed to assess intermittent, high-intensity fitness have yet to be created. A few soccer-specific fitness testing procedures have been accepted by the nonprofessional soccer community, but most lack sound research design or validity. These tests include, but are not limited to, maximal aerobic power and functional tests developed to correlate directly with soccer match-specific activity (1, 8, 9, 16, 19, 20, 25). To assess team match fitness, the current study used a new tool previously used only at the elite level—the Loughborough Intermittent Shuttle Test (LIST). This test has been validated to be an accurate predictor of soccer performance in elite soccer players. However, to apply this test to our population, the test was modified to more closely simulate “nonelite” match requirements. Individual indices of intermittent, high-intensity fitness were assessed using measurement techniques common at the elite level (2, 24, 26).

Methods

Experimental Approach to the Problem

Thirty-four female high school soccer players were tested prior to a competitive 10-week season and again at the end of the same competitive season. Approval was obtained from the Internal Review Board for research with human subjects at the University of Montana, and informed consent was given by the players and their parents or guardians.

The tests included an abridged version of the LIST (19), hydrostatic weighing using estimated residual volumes, vertical jump, 20-m sprint, and 30-second Wingate test. The individual variables were chosen to coincide with the most commonly occurring soccer-specific fitness tests found throughout the literature. The abridged LIST was performed as a group test by each team to measure power endurance. Tests estimating body density (hydrostatic weighing) and anaerobic power outputs (vertical jump, 20-m sprint, and 30-second Wingate test) were conducted individually.

Subjects

Subjects were members of varsity teams from 2 similar-sized high schools in Missoula, Montana. Neither team participated in preseason conditioning programs, and the athletes were of similar caliber (assessed from the previous season's similar conference records). The experimental intervention team (experimental group: EG) implemented a series of resistive training and plyometric exercises throughout the 10-week high school season. The resistive training consisted of 2 sessions a week, taking place just prior to practices. Each session required approximately 30 minutes. Three sets of 4 lower-body free weights exercises (squats, leg extensions, leg curls, and calf raise) were completed each strength session. Pre- and post-season training 1-repetition maximum (1RM) was es-

timated from 10–15RM efforts using equations developed by Landers (14). The resistance training focus varied from explosive power sets to slower muscle-building activities. The investigator supervised all weight-training sessions.

Testing Procedures

Plyometric sessions lasted approximately 10–15 minutes and were conducted 3 times per week during practice. The plyometric sessions were scheduled on alternate days from the strength-training sessions. Explosive jumping (box jumps), quickness (timed jumping, rejumping), and power endurance (skipping for height and distance) were stressed during the plyometric workouts. Typical training sessions used either the width of the soccer field or the 18-yard box. The exercises progressively increased in quantity and intensity as the fitness level improved throughout the season (3 to 5 sets). In addition, the jumping heights were increased. All plyometric sessions were conducted after skill-specific soccer sessions were completed. Individual data on both strength and plyometric performance were recorded throughout the experimental period. The investigator conducted all training sessions. A summary of the training program for the EG is given in the Appendix.

The control team (control group: CG) did not partake in a regimented resistive training or plyometrics program. The training program for the CG consisted of distance and endurance training 1–2 times per week, plus regular soccer practices and matches. Additional shuttle runs were implemented as the coach assessed fitness throughout the season. Resistive training was eliminated from the CG's training program.

Testing Dates

Baseline testing was scheduled to coincide with the beginning of the 10-week competitive high school soccer season. Posttesting was scheduled within 1 week after the final game of the season.

Tests

Abridged Version of LIST. The abridged version of the LIST was given to all participants pre- and postseason and was completed prior to the individual tests. The test was conducted at indoor gymnasiums on the campus of the University of Montana, Missoula, MT. The running surface was composed of wood and was cleaned before the test was conducted. Two days prior to the tests, the teams were given standard pretest instructions for diet, exercise, and clothing. Upon arrival to the gymnasium, the athletes warmed up and stretched. Players were then familiarized with the testing procedure via verbal explanation and a practice trial. All questions were answered prior to the start of the test.

The abridged version of the LIST consisted of 2 parts: Part A, a soccer match simulation period, and

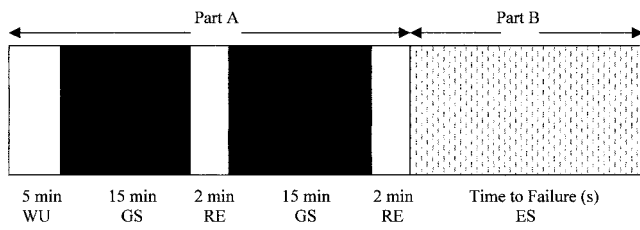
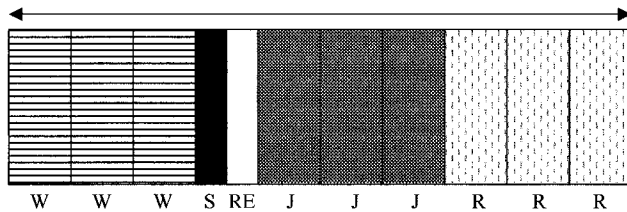


Figure 1. The abridged Loughborough Intermittent Shuttle Test (LIST), consisting of 2 15-minute game simulation periods of variable-intensity shuttle running over 16 m and concluding with a fatigue phase. WU = warm-up; GS = game simulation; ES = exhaustive stage; RE = rest.



Repeated 9 times per 15 minute bout

W: walk
S: sprint
RE: rest
J: jog
R: run

Figure 2. The cycle of laps during the abridged Loughborough Intermittent Shuttle Test (LIST), repeated 9 times during each 15-minute game simulation. W = walk; S = sprint; RE = rest; J = jog; R = run.

Part B, an exhaustive exercise stage. The test used a 16-m distance over which the players walked, jogged, ran, and sprinted laps in a specific order and at specific speeds. The lap speeds were estimated from trial times recorded in a pilot study. The speeds were dictated by an audio signal recorded on audiotape. Audio signals indicated the start and end times for each lap. The pattern of exercise for the game simulation was as follows (Figure 1).

Part A. Two 15-minute game simulation periods of variable-intensity shuttle running over 16 m, separated by a 2-minute rest. The cycle of laps (see Figure 2) was repeated 9 times during each 15-minute game simulation, in the following order (times shown are the lap time for 16 m): (a) 3 laps at walking pace (18.5 s/lap^{-1}), (b) 1 lap at maximal running speed (4.1 s/lap^{-1}), (c) 4 seconds recovery, (d) 3 laps at running speed corresponding to 55% of individual estimated maximal sprint speed (7.5 s/lap^{-1}), and (e) 3 laps at running speed corresponding to 95% of individual estimated maximal sprint speed (5.2 s/lap^{-1}).

Part B. The exhaustion exercise stage was designed to become progressively more difficult until volitional exhaustion was reached. The mode consisted of alternating jogging and running (Figure 2). Initial times were 8 seconds jogging in 1 direction for 16 m, and

Table 1. Descriptive data at baseline testing.*

Subject data	Control group (<i>n</i> = 17)	Experimental group (<i>n</i> = 17)
Age	16.27 ± 1.38	16.49 ± 0.91
Height (cm)	166.7 ± 4.71	167.42 ± 4.64
Body weight (kg)	58.00 ± 7.23	61.46 ± 9.43
FFM (kg)	48.96 ± 4.63	49.33 ± 6.37
FM (kg)	9.17 ± 4.11	12.13 ± 4.66

* Data presented as mean $\pm$ standard deviation. FFM = fat-free mass; FM = fat mass.

then returning with a 5.7-second run. Subjects were allowed 2-second rest intervals between laps to decelerate and reverse direction. Jogging and running times were decreased by 0.1 s/lap^{-1} every minute until volitional exhaustion was reached or the subject was unable to meet the required time on 2 consecutive running laps. Time to volitional exhaustion was recorded. The test was administered separately to each team, and the results were recorded individually for each player.

Individual Tests

All individual tests were conducted both pre- and postseason at least 1 day after completion of the abridged LIST. Details of the individual tests are discussed below.

Descriptive Variables. The subject's height without shoes was determined to the nearest 0.1 cm with the use of a stadiometer (Narragansett Machine Company, Providence, RI). The subject's weight, wearing only a bathing suit, was recorded to the nearest 0.1 kg with the use of a calibrated electronic Befour weight scale (Cedarburg, WI). Age was recorded to the nearest year at the start of the study (Table 1).

Body Composition. Body composition was estimated using hydrodensitometry. Underwater weight was measured using an electronic scale (Exertech Body Density Measurement Systems, Dresbach, MN), which was calibrated prior to each measurement. The subjects were allowed as many trials as necessary for a consistent underwater weight measurement until 2 trials were recorded within 0.01 kg. Underwater weights were recorded to the nearest 0.001 kg. Residual lung volume was estimated using equations derived from Forsyth et al. (10). The same residual volume was used for both pre- and posttesting measurements. Body density was calculated from underwater weight after adjustments for water temperature, residual volume, and 200 ml gastrointestinal gas (5). Percentage body fat was estimated using age- and sex-dependent equations of Lohman (17) and Siri (23). Percentage body fat was used to estimate fat-free mass (FFM) and fat mass (FM).

Vertical Jump. Vertical jump was measured using a

Table 2. Change in mass and body composition with training.†

Variable	Group	Pretest	Posttest	Change pre- to posttest
Body weight (kg)	CG	58.00 ± 7.23	58.14 ± 7.65	-0.04 ± 1.82
	EG	61.46 ± 9.43	61.21 ± 9.26	-0.25 ± 1.82
FFM (kg)	CG	48.96 ± 4.63	49.56 ± 4.88	0.59 ± 1.84
	EG	49.33 ± 6.37	50.48 ± 6.92*	1.14 ± 1.22
FM (kg)	CG	9.17 ± 4.11	8.54 ± 4.07	-0.63 ± 1.61
	EG	12.13 ± 4.66	10.73 ± 4.26*	-1.40 ± 1.47

† Data presented as mean ± standard deviation. CG = control group; EG = experimental group; FFM = fat-free mass; FM = fat mass.

* $p < 0.05$ pre- to postseason within-group testing.

wall tape. Subjects were instructed to reach above their head within normal extension limits to record standing reach. The same initial reach was used in both pre- and postseason measurements. The subjects were then allowed 3 trial jumps to obtain their maximal vertical jump (in centimeters). Subjects were allowed arm movement but not a preliminary step.

Twenty-Meter Sprint. The 20-m sprint was recorded using an Alge-Sports electronic timer, Model S3 (ALGE Electronic Timing, Lustenau, Austria) set at a height of 25 cm. The trial was conducted outside on a concrete sidewalk. All subjects were required to warm up 10 minutes prior to the sprint and were allowed 3 separate trials. A distance of 30 m was marked at 0, 10, and 30 m with a tape. The timing lights were placed at 10 and 30 m. For each trial date, the lights were carefully aligned to the exact position of prior trials using permanent marks on the pavement; subsequent measuring and vertical alignment used a carpenter's level. Subjects accelerated for 10 m before breaking the first light beam and starting the clock. The final elapsed time was recorded at 30 m. The fastest time of the 3 trials was accepted as the subject's maximum sprint speed.

Thirty-Second Wingate Test. Subjects were allowed 10 minutes warm-up at 70 revolutions per minute on a Monarch 824 weighted cycle ergometer (Varberg, Sweden). During the warm-up period, subjects were briefed on the dynamics of the test and what was expected of them. The standard 30-second Wingate protocol was followed with resistance set at 7.5% of the subject's body weight rounded to the nearest 0.5 kg. Once the subject verbally acknowledged she was ready, the test was initiated. Peak, average, and minimum power (in watts), as well as percentage decline ($[\text{peak} - \text{minimum}]/\text{peak}$) were measured and adjusted for flywheel inertia using a computerized recording system (SMI, St. Cloud, MN).

Statistical Analyses

To evaluate within- and between-group differences in changes (pre- to postseason) in the time to failure on

the LIST test, a 2-way, mixed-design analysis of covariance (ANCOVA; 1 within, 1 between), covaried for pretraining time to failure on the LIST test, was used. The ANCOVA analysis was used for the LIST data because of the large pretraining difference in time to failure on the LIST test between the 2 groups. The decision to use an ANCOVA (vs. analysis of variance [ANOVA]), which also showed significant group differences) was made post hoc to better adjust for baseline differences. It should be noted that the subjects in this analysis were not randomly assigned, reducing the power of both the ANCOVA and ANOVA analyses. Power and body composition measurements were assessed using a 2-way, mixed-design ANOVA (1 within, 1 between) with a priori planned comparisons to evaluate differences between groups. Mean change pre- to postseason was assessed using a 1-tailed independent *t*-test to determine differences in group-specific responses. A priori significance was set at $p \leq 0.05$.

Results

Table 2 includes descriptive body composition data pre- to posttraining portrayed as group means (EG and CG). No significant differences between groups were observed for age, height, or weight at baseline or pre- to postseason (see Table 1). Significant differences were observed in the EG for changes of FM (-1.40 ± 1.47 , $p = 0.004$) and FFM (1.14 ± 1.22 , $p = 0.0008$).

Table 3 shows pre- and posttraining means and standard deviations for the LIST and for each power variable tested by group. Significant differences for the abridged LIST were evaluated using an ANCOVA ($p = 0.02$) covaried for pretraining LIST data. Significant pre- to posttraining differences between groups for the other power variables were evaluated using a 2-way, mixed ANOVA (1 within, 1 between) with a priori planned comparisons.

LIST

The CG showed a nonsignificant improvement in overall time to failure during the LIST test (50.00 seconds

Table 3. Pre- and postseason data for the vertical jump, 20-m sprint, and Wingate test.

Variable	Group	Pretest	Posttest	Change (posttest – pretest)
LIST (seconds to failure)	CG	1,064.00 ± 195.15	1,115.00 ± 157.51	50.00 ± 176.37
	EG	646.00 ± 167.47	1,040.00 ± 157.33	394 ± 123.59*
Vertical jump (cm)	CG	36.46 ± 3.68	39.19 ± 4.45	0.73 ± 3.37
	EG	37.65 ± 4.77	39.37 ± 4.69	1.72 ± 3.83
20-m sprint (s)	CG	2.89 ± 0.13	2.85 ± 0.13	–0.04 ± 0.09
	EG	3.00 ± 0.15	2.90 ± 0.13	–0.10 ± 0.10*
Peak power (kg·m·min ^{–1})	CG	9.59 ± 0.92	9.78 ± 1.36	0.19 ± 1.01
	EG	10.36 ± 2.38	10.68 ± 2.20	0.32 ± 3.61
Average power (kg·m·min ^{–1})	CG	7.76 ± 0.60	7.73 ± 0.78	–0.03 ± 0.51
	EG	7.27 ± 0.49	7.37 ± 0.64	0.10 ± 0.58
Minimum power (kg·m·min ^{–1})	CG	5.80 ± 0.50	5.65 ± 0.74	–0.14 ± 0.58
	EG	4.86 ± 0.65	4.91 ± 0.97	0.05 ± 0.93
Percentage decline	CG	39.23 ± 6.00	42.00 ± 8.00	0.02 ± 0.06
	EG	51.00 ± 12.45	52.45 ± 12.00	0.02 ± 0.16

† Data presented as mean ± standard deviation. LIST = Loughborough Intermittent Shuttle Test; CG = control group; EG = experimental group.

* $p < 0.05$: analysis of covariance change between groups pre- to postseason for abridged LIST test; significant differences within groups using planned comparisons pre- to postseason for all variables except abridged LIST test.

± 176.37 seconds), whereas the EG improved significantly (394.00 seconds ± 123.59 seconds) more than the CG. The 344.00-second greater improvement by the EG than the CG on the LIST was significant ($p = 0.02$), even after adjusting for baseline values (Figure 3).

Vertical Jump

Both the EG and CG improvements pre- to postseason were nonsignificant (1.72 cm ± 3.83 cm, $p = 0.06$; 0.73 cm ± 3.37 cm, $p = 0.46$).

Twenty-Meter Sprint

Significant improvements in sprint times were observed pre- to postseason for EG (–0.10 seconds ± 0.10 seconds, $p = 0.0003$). The CG's reduction in sprint time pre- to postseason was not significant (–0.04 seconds ± 0.09 seconds, $p = 0.15$).

Wingate Cycle, Leg Power Test

No significant within- or between-group differences were observed pre- to posttraining in peak power, average power, minimum power, and percentage power decline in the Wingate test data. This nonsignificant difference was consistent whether data were evaluated in absolute or body weight-adjusted values.

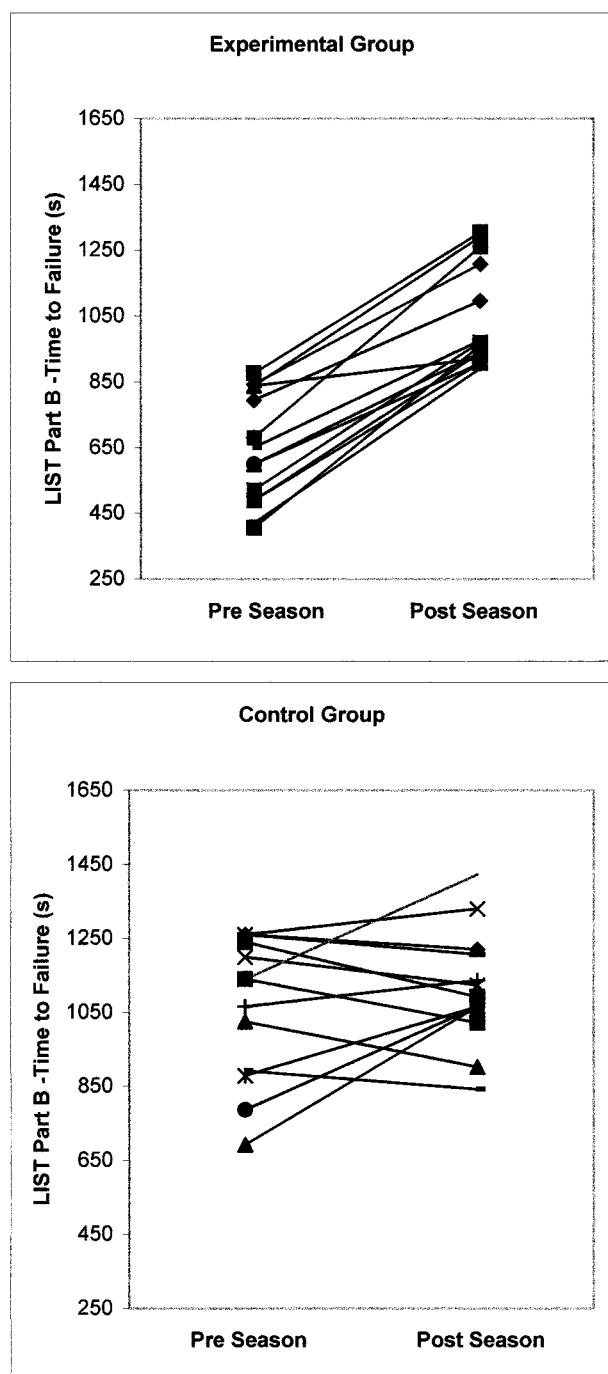
Body Composition

No significant differences between groups were observed in the change in body weight pre- to posttraining. Significant pre- to postseason changes were observed in the EG for FM (–1.40 ± 1.47, $p = 0.004$) and FFM (1.14 ± 1.22, $p = 0.0008$). No significant changes were observed pre- to postseason for FM and FFM values in the CG.

Discussion

The present findings indicate that soccer-specific power endurance training with young women enhances intermittent, high-intensity fitness. Incorporating power endurance training into women's soccer programs while reducing the focus of aerobic training improved fitness variables related to soccer match performance. In the current study, the EG's significant posttraining increase in time to failure on the abridged LIST (+394.00 seconds ± 123.59 seconds) as compared with the CG's posttraining increase (+50.00 seconds ± 176.37 seconds) was significant ($p = 0.02$) between groups. Initial preseason differences were taken into account through statistical procedures (ANCOVA). The CG's large standard deviation values may be attributed to the different individual responses to training (Figure 3). These data also indicate that adjusting the soccer-specific LIST, validated in studies using elite athletes, may be an effective technique in evaluating soccer-specific fitness in young women when proper adjustments are made for the target population. However, further study of the validity and reliability of the abridged LIST is needed.

To the investigators' knowledge, this study was the first to abridge the LIST. Therefore, comparative results were absent and true validity difficult to obtain. However, during the course of the study, the coaches of both teams were in general agreement that those athletes who had scored higher on the LIST during the preseason testing were better able to maintain their field effectiveness late in the match. In postseason discussions with the coaches, there was again general



made that the Wingate was an accurate indicator of lower-body power output as it relates to soccer performance. However, in the present study, no pre- to postseason improvements in anaerobic capacity were observed in either group when assessed with a 30-second Wingate cycle test. The lack of change in power assessed by the Wingate cycle test, differing from the 20-m sprint and vertical jump tests, which both showed improvement, may be attributed to the lack of specific cycle training. These findings indicate that the Wingate cycle test may not be appropriate to assess improvements throughout the course of a 10-week season for young women. This may be due to the difference in specificity of the soccer training compared with the Wingate cycle test, or possibly to the lack of additional training to improve the glycolytic system over what these athletes normally do in training. Using a 20-m sprint or vertical jump measurement may be more applicable when determining anaerobic power specific to young female soccer players. Sprint and vertical jump measurements may also be more practical for youth coaches, as they require little expense or equipment.

Soccer is a complex sport requiring a combination of individual skill, power, endurance, speed, and experience, along with the ability for a team to function cohesively as a unit. This research attempted to measure only the fitness variables of 2 similar women's high school soccer teams. Discussed was the possibility that the improved performance of the EG was related to their improved fitness. However, since coaching, team play, and skill were not controlled, it remains difficult to link changes in fitness to changes in performance.

Designing research-based soccer training protocols for young women is difficult, because there is limited research and few validated sport-specific measures to evaluate adolescent soccer-specific training. The abridged LIST protocol designed for this study, though not validated, appears to be a useful and beneficial means for assessing young women's soccer match fitness. Although match performance was not directly measured in the current study, the season-long results of the 2 teams competing against the same opponents, in comparison with the previous season, further suggest that the abridged LIST may be somewhat related to match performance (which, one assumes, is related somewhat to fitness). Our data would support that the differences in improvements in speed and soccer-specific power endurance (and possibly performance) were most likely a result of the differences in training protocols throughout the 10-week season, with the EG showing significant improvements, whereas no improvement was observed in the CG. As indicated by the EG's coach, the team was less fatigued not only throughout the match, but progressively showed an increased level of fitness throughout

the season. The EG's improved fitness may also have led to an increase in wins over the previous season, as well as an increase in goals scored and a decrease in goals allowed. The CG's coach, although reporting a gradual increase in fitness throughout the season, experienced player fatigue as individual matches progressed and little change in outcome performance between the research year and the prior year.

Practical Applications

The implementation of a soccer-specific intermittent, high-intensity exercise protocol during the high school soccer season appears to be beneficial in improving match fitness in young women. Abridging the LIST to fit the needs and abilities of young women appears to be useful. However, the abridged LIST still requires rigid validation, and the results of this study should be viewed in that context, which will ideally stimulate further research.

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Appendix. Phylometric/Strength Practice Program.*

Prac- day	Strength/ power	Notes
Mon	Plyos	"Do or die" prone: 7 players $\times$ 3 sets; butt kicks, single-legged, PA width; skipping, vertical height, 15 each leg; side to side hops, 20 each direction, 2 feet; one-legged squats, 5 each leg
Tue	Strength	1 set 6–12RM testing/instruction
Wed	Plyos	"Do or die" crouch: 5 players $\times$ 2 sets; butt kicks, double-legged, GB width; vertical jumping, max height, 2 at a time, 8 $\times$ 2 jumps; clock hops, 2 $\times$ around, 2.5-ft radius; one-legged vertical jumps, 5 each leg
Thu	Strength	2 sets, 12–15RM, 3-min rest between sets, move slowly. set-rep
Fri	Plyos	Depth jumps, 6 from 6 inches; "do or die" kneel (arms and legs): 3 players $\times$ 3 sets; two-legged hops, distance, PA width; "hang time" jumps, 15 seconds
Mon	Plyos	Skipping, distance, PA width; side cross-over running, PA width $\times$ 2; side to side hops, 20 each direction, 3 ft; one-legged vertical jumps, 6 each leg
Tue	Strength	2 sets, 12–15RM, 3-min rest between sets, move explosively. set-rep
Wed	Plyos	"Do or die" prone: team $\times$ 3 sets; butt kicks, singled-legged, PA width; skipping, vertical height, 18 each leg; side to side hops, 20 each direction, 3 ft; one-legged squats, 7 each leg
Thu	Strength	2 sets, 12–15RM, 3-min between sets, move explosively. set-rep
Fri	Plyos	"Do or die" crouch: 5 players $\times$ 2 sets; vertical jumping, max height, 2 at a time, 8 $\times$ 2 jumps; clock hops, 3 $\times$ around, 2.5-ft radius
Mon	Plyos	Depth jumps, 8 from 8–10 inches; "do or die" kneel (arms and legs): 5 players $\times$ 3 sets; two-legged hops, distance, PA width; "hang time" jumps, 15 seconds; one-legged squats, 6 each leg
Tue	Strength	2 sets, 8–10RM, 3–4-min rest between sets, move explosively. set-rep
Wed	Plyos	Skipping, distance, PA width; side cross-over running, PA width $\times$ 2; side to side hops, 20 each direction, 3 feet; one-legged vertical jumps, 6 each leg
Fri	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly. set-rep
Mon	Plyos	"Do or die" prone: team $\times$ 3 sets; butt kicks, single-legged, PA width; skipping, vertical height, 22 each leg; side to side hops, 30 each direction, 3 ft; one-legged squats, 10 each leg
Tue	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly. set-rep
Wed	Plyos	"Do or die" crouch: 5 players $\times$ 2 sets; vertical jumping, max height, 2 at a time, 8 $\times$ 2 jumps; speed skater jumps, 10 each direction
Mon	Plyos	Depth jumps, 8 from 8–10 inches; "do or die" kneel (arms and legs): 5 players $\times$ 3 sets; two-legged hops, distance, PA width; "hang time" jumps, 10 seconds max effort; one-legged squats, 6 each leg
Tue	Strength	3 sets, 1st: 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly, 1-min rest, 3rd 20RM light weights, explosive. set-rep
Wed	Plyos	Skipping, distance, PA width $\times$ 2; side cross-over running, PA width $\times$ 2; side to side hops, 20 each direction, 3 ft; one-legged vertical jumps, 6 each leg

Appendix. Continued.

Prac- tice day	Strength/ power	Notes
Thu	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly
Fri	Plyos	“Do or die” crouch: 7 × 3 sets; butt kicks, single-legged, PA width; skipping, vertical height, 22 each leg; side to side hops, 30 each direction, 3 feet; depth jumps, 12–14 inches × 6
Mon	Plyos	“Do or die” crouch: 7 players × 2 sets; speed skater jumps, 10 each direction; vertical jumping, max height, 2 at a time, 8 × 2 jumps; clock hops, 2 × around, 2.5-ft radius; one-legged vertical jumps, 5 each leg
Tue	Strength	3 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly, 1-min rest, 3rd 20RM light weights, explosive. set-rep
Wed	Plyos	Depth jumps, 8 from 8–10 inches; “do or die” kneel (arms and legs): 5 players × 3 sets; two-legged hops, distance, PA width; “hang time” jumps, 10 seconds max effort; one-legged squats, 6 each leg
Thu	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly
Fri	Plyos	Skipping, distance, PA width × 1; side cross-over running, PA width × 2; side to side hops, 20 each direction, 3 ft; depth jumps, 6–8 inches × 5
Mon	Plyos	Depth jumps, 12–14 inches × 8; “do or die” crouch: 7 × 3 sets; butt kicks, single-legged, PA width × 2; skipping, vertical height, 15 each leg hard effort; side to side hops, 10 with maximum effort (distance) each direction
Tue	Strength	3 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly, 1-min rest, 3rd 20RM light weights, explosive. set-rep
Wed	Plyos	“Do or die” crouch: 7 players × 2 sets; speed skater jumps, 10 each direction; vertical jumping, max height, 2 at a time, 8 × 2 jumps; clock hops, 2 × around, 2.5-ft radius; one-legged vertical jumps, 5 each leg
Thu	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 4–6RM slowly
Fri	Plyos	Skipping, distance, PA width × 1; side cross-over running, PA width × 2; side to side hops, 20 each direction, 3 ft; depth jumps, 6–8 inches × 5
Wed	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 10–12RM explosively. set-rep
Thu	Plyos	Depth jumps, 8 from 8–10 inches; “do or die” kneel (arms and legs): 5 players × 3 sets; two-legged hops, distance, PA width; “hang time” jumps, 10 seconds max effort; one-legged squats, 6 each leg
Sat	Plyos	“Do or die” prone: team × 3 sets; butt kicks, single-legged, PA width × 2; skipping, vertical height, 10 each leg with maximal effort; side to side hops, 15 each direction, maximum distance; one-legged squats, 10 each leg
Wed	Plyos	“Do or die” crouch: 7 players × 2 sets; speed skater jumps, 10 each direction; vertical jumping, max height, 2 at a time, 8 × 2 jumps; clock hops, 2 × around, 2.5-ft radius; one-legged vertical jumps, 5 each leg
Thu	Strength	2 sets, 1st 8–10RM explosively, 3–4-min rest, 2nd 10–12RM explosively. set-rep
Fri	Plyos	Depth jumps, 8 from 8–10 inches; “do or die” kneel (arms and legs): 5 players × 3 sets; two-legged hops, distance, PA width; “hang time” jumps, 10 seconds max effort; one-legged squats, 6 each leg
Sat	Strength	1 set, 8–10RM explosively
Mon	Plyos	Skipping, distance, PA width × 1; side cross-over running, PA width × 2; side to side hops, 20 each direction, 3 ft; depth jumps, 6–8 inches × 5

* Plyos = plyometrics; PA = penalty area; GB = goal box; RM = repetition maximum; set-rep = repeat set.